

Flood Inundation Mapping of the Narayani River Basin, Nepal, Using HEC-HMS and HEC-RAS 2D Hydrodynamic Modeling

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ABSTRACT

Flood inundation mapping is a critical component of disaster risk reduction in river basins prone to seasonal and extreme flood events. The Narayani River basin, one of Nepal's largest and most flood-prone river systems, experiences severe inundation during the monsoon season, threatening lives, infrastructure, and agricultural livelihoods across the Terai plains. This study presents a comprehensive flood inundation mapping framework for the Narayani basin using an integrated hydrological and hydrodynamic modeling approach.

A multi-step methodology was applied: watershed delineation in ArcGIS Pro; rainfall frequency analysis using a Log-Pearson Type III distribution fitted to CHIRPS (1990-2024) rainfall data in EasyFit; rainfall-runoff simulation in HEC-HMS using the SCS Curve Number loss method and SCS Unit Hydrograph transform; and 2D unsteady-flow hydraulic simulation in HEC-RAS. Land cover was classified from ESA WorldCover 2021 in Google Earth Engine, and hydrologic soil groups were derived from OpenLandMap soil texture data. The study area comprised a 23,425.8 km² watershed delineated into 9 sub-basins linked by 4 routing reaches.

Flood frequency analysis yielded design storm depths from 93.66 mm (2-year) to 236.45 mm (100-year return period). HEC-HMS simulations produced peak discharges at the watershed outlet ranging from 1,374.9 m³/s (2-year) to 8,551.5 m³/s (100-year). HEC-RAS 2D modeling was applied to a 37.6 km reach with a 4 km buffer floodplain, incorporating spatially distributed Manning's roughness derived from land cover classification. The resulting maps delineate flood extent, depth, and velocity for six return periods, offering critical inputs for flood risk assessment, land-use planning, and early-warning system development in the Narayani floodplain.

Keywords: flood inundation mapping; HEC-HMS; HEC-RAS 2D; SCS curve number; Log-Pearson III; Narayani River; Nepal; disaster risk reduction; remote sensing; GIS

1. INTRODUCTION

Nepal is one of the most disaster-prone countries in South Asia, with floods ranking among the most frequent and devastating natural hazards. The Narayani River, known as the Gandaki in India, is one of Nepal's three major river systems and drains a significant portion of the central Himalayan region. Its catchment encompasses diverse physiographic zones ranging from the high Himalayas to the low-lying Terai plains, creating complex hydrological dynamics that result in severe flood events during the monsoon season (June-September).

Flood inundation mapping is an essential tool for disaster risk management, enabling planners, policymakers, and emergency responders to understand the spatial extent, depth, and frequency of flood events. Traditional mapping approaches based on historical records are insufficient given the non-stationary nature of flood regimes under changing climate conditions. Physics-based hydrological and hydrodynamic modeling provides a robust framework for estimating flood inundation under various return-period scenarios, an approach that has been applied successfully to comparable coupled 1D/2D problems in South Asian floodplains (Patel et al., 2017).

This study integrates rainfall-runoff modeling (HEC-HMS; USACE, 2016) with 2D hydrodynamic flood routing (HEC-RAS; Brunner, 2016) to produce flood inundation maps for six return periods (2, 5, 10, 25, 50, and 100 years) for a critical 37.6 km reach of the Narayani River. The integrated framework leverages freely available remote sensing data, including CHIRPS rainfall (Funk et al., 2015), ESA WorldCover land cover (ESA, 2021), and the ALOS PALSAR digital elevation model, building on the growing availability of global topographic and hydrographic datasets for data-scarce basins (Yamazaki et al., 2019). This data-driven design makes the framework replicable across the data-scarce environments typical of developing nations.

1.1 Objectives

- To delineate the Narayani River watershed and characterize its hydrological parameters
- To perform rainfall frequency analysis using the Log-Pearson III distribution
- To derive spatially distributed SCS Curve Numbers from land cover and soil data
- To simulate rainfall-runoff processes using HEC-HMS for six return periods

- To perform 2D unsteady-flow hydraulic simulation using HEC-RAS
- To produce flood inundation maps showing flood extent, depth, and velocity

2. STUDY AREA

The Narayani River basin is in central Nepal, bounded approximately between 82.5°E-85.5°E longitude and 26.5°N-29.5°N latitude. The basin drains an area of approximately 27,335 km² within Nepal before crossing into India, where it is known as the Gandaki River, a major tributary of the Ganges. The watershed encompasses three major physiographic zones: the High Himalayas (>5,000 m), the Middle Hills (500-5,000 m), and the Terai plains (<200 m).

- To support flood risk assessment and disaster risk reduction planning in the Narayani basin

For this study, the hydrological modeling domain covers a watershed delineated to a total area of 23,425.8 km², comprising 9 sub-basins ranging from 66.1 km² (Sub-basin 9) to 7,636.4 km² (Sub-basin 4). The hydraulic modeling domain focuses on a 37.6 km study reach extending from the gorge exit near Devghat to downstream of Narayanghat, where the river transitions from a confined mountain gorge to the expansive Terai floodplain. This transition zone is of critical importance for flood hazard assessment, as it represents the region where floodwaters spread laterally onto the floodplain, threatening the densely populated Chitwan district.

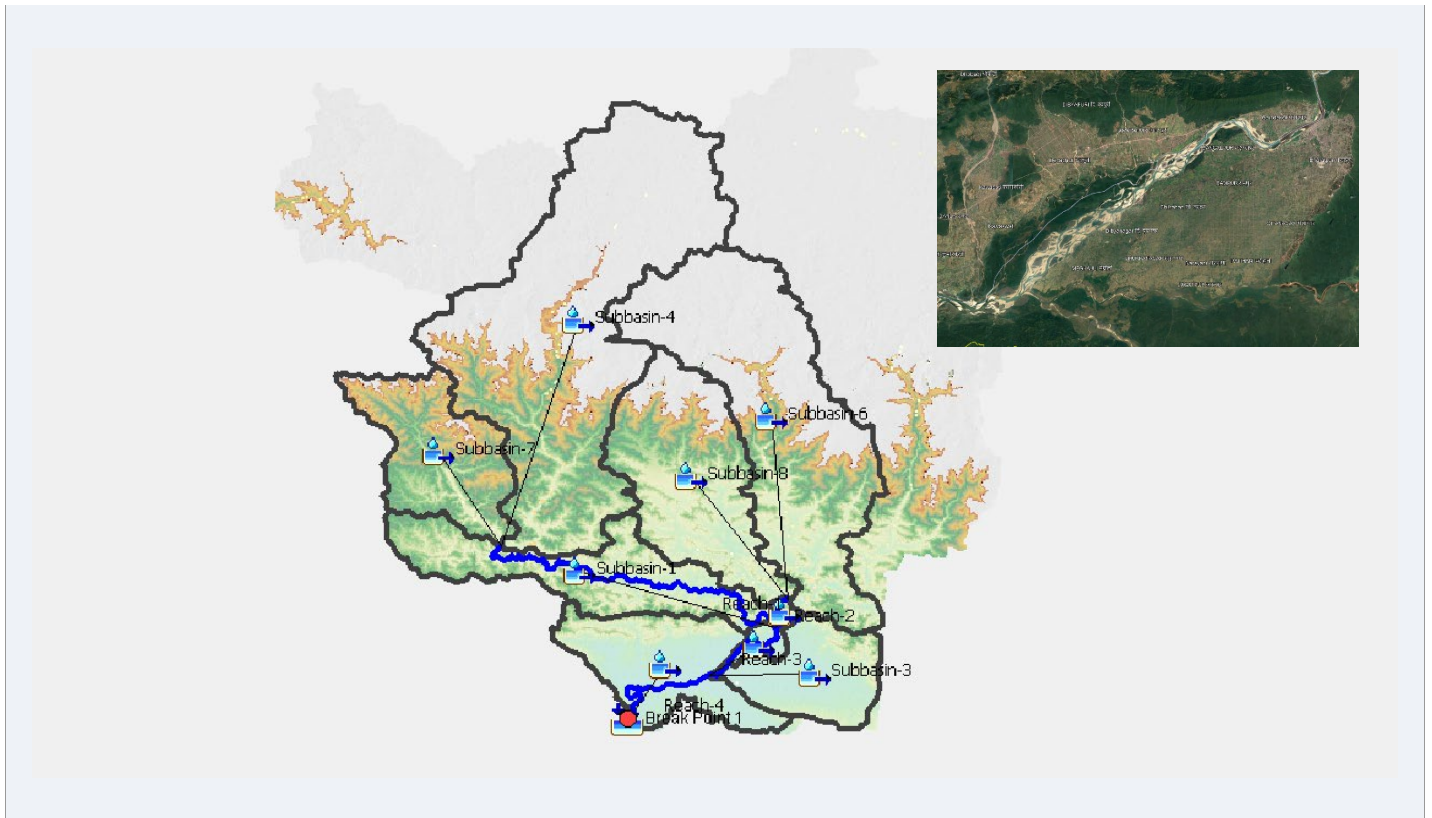


Figure 1. Location map of the Narayani River basin showing the watershed boundary, sub-basin delineation, river network, study reach extent, physiographic zones, and major settlements.

Table 1. Key characteristics of the Narayani River study area.

Parameter	Value
Total watershed area	23,425.8 km ²
Number of sub-basins	9
Number of routing reaches	4
Study reach length	37.6 km
Study reach buffer width	4 km (each side)
Elevation range	76 m – >8,000 m
Major river	Narayani (Gandaki)
Administrative location	Gandaki Province, Nepal
Climate zone	Subtropical to alpine
Monsoon season	June – September

3. DATA AND SOFTWARE

3.1 Data Sources

All datasets used in this study are freely available global or regional remote sensing products, summarized in Table 2, chosen specifically to support replicability in data-scarce Himalayan catchments.

Table 2. Data sources used in the study. *ERA5 discharge was used for reference only; rainfall-based HMS simulation was adopted as the primary approach due to ERA5 overestimation in Himalayan basins.

Data Type	Source	Resolution / Period	Use
Rainfall	CHIRPS v2.0 (Funk et al., 2015)	Daily, 1990-2024	Frequency analysis
DEM	ALOS PALSAR	12.5 m	Terrain modeling
Land cover	ESA WorldCover 2021 (ESA, 2021)	10 m	CN & Manning's n
Soil texture	OpenLandMap	250 m	HSG derivation
Discharge	ERA5 reanalysis*	Monthly	Validation reference
Imagery	Google Satellite	Various	Visual reference
River network	ArcGIS Pro	—	Watershed delineation

3.2 Software

Table 3. Software tools used in the study.

Software	Version	Purpose
ArcGIS Pro	3	Watershed delineation, GIS analysis, map production
Google Earth Engine	Cloud	LULC extraction, soil processing, CN computation
EasyFit	5.5	Log-Pearson III flood frequency analysis
HEC-HMS (USACE, 2016)	4.13	Rainfall-runoff modeling, hydrograph generation
HEC-RAS (Brunner, 2016)	7.0.1	2D unsteady-flow hydraulic simulation
Microsoft Excel	2021	SCS CN, Tc, and lag-time calculations

4. METHODOLOGY

The methodology follows an integrated hydrological-hydrodynamic modeling framework consisting of five sequential phases, as illustrated in Figure 2.

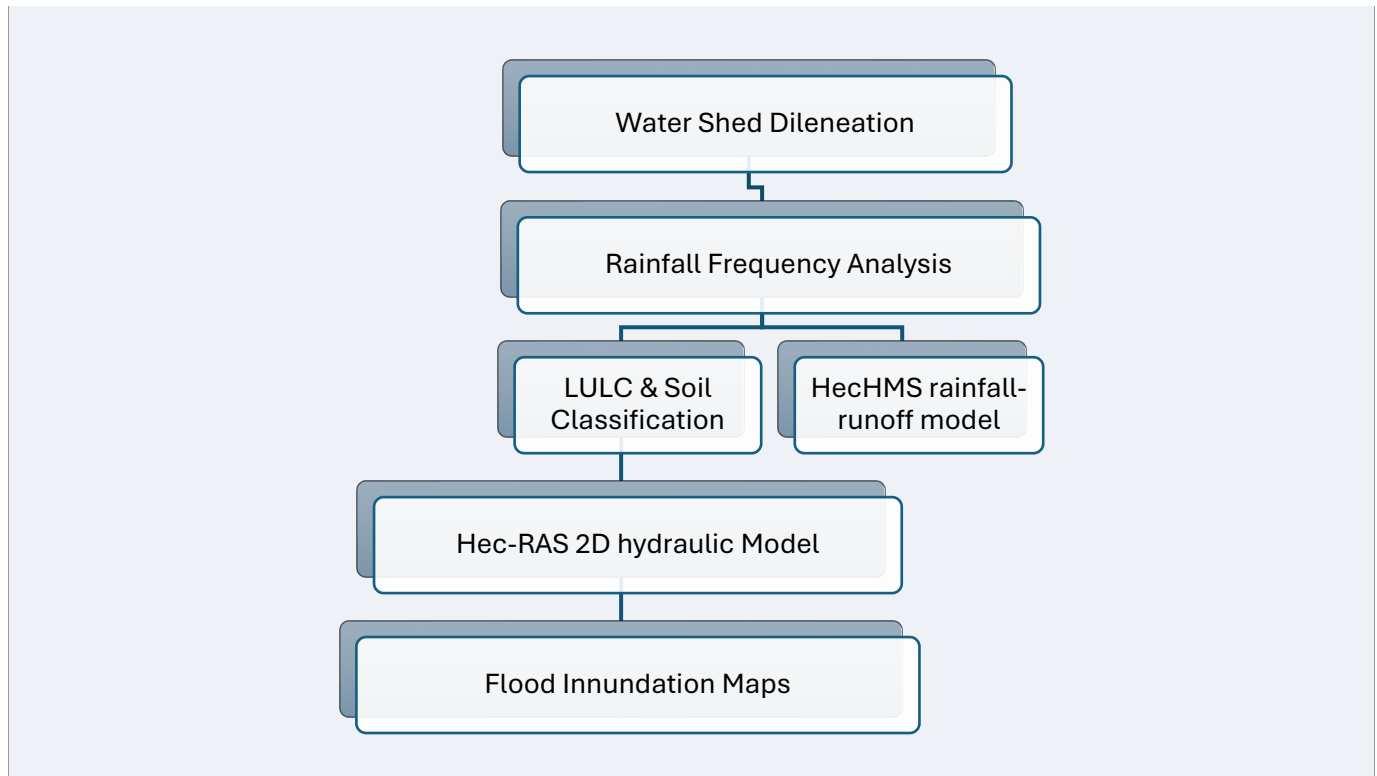


Figure 2. Methodology flowchart showing the integrated modeling framework from data collection to flood inundation map production, with all major steps and data flows.

4.1 Watershed Delineation

Watershed delineation was performed in ArcGIS Pro using the ArcHydro extension. The ALOS PALSAR DEM at 12.5 m resolution was used as the primary terrain input. Preprocessing steps included:

- DEM mosaicking and projection to UTM Zone 45N (EPSG:32645)
- Depression filling to remove sinks and ensure continuous flow routing
- Flow direction computation using the D8 algorithm
- Flow accumulation grid generation

- Stream network delineation using a flow accumulation threshold
- Watershed boundary and sub-basin delineation using pour points
- Computation of basin morphometric parameters, including longest flow path, basin slope, and drainage density

The delineation produced 9 sub-basins covering a total area of 23,425.8 km², connected by 4 routing reaches. Sub-basin morphometric parameters were extracted and used as inputs for the SCS Curve Number method and time-of-concentration calculations.

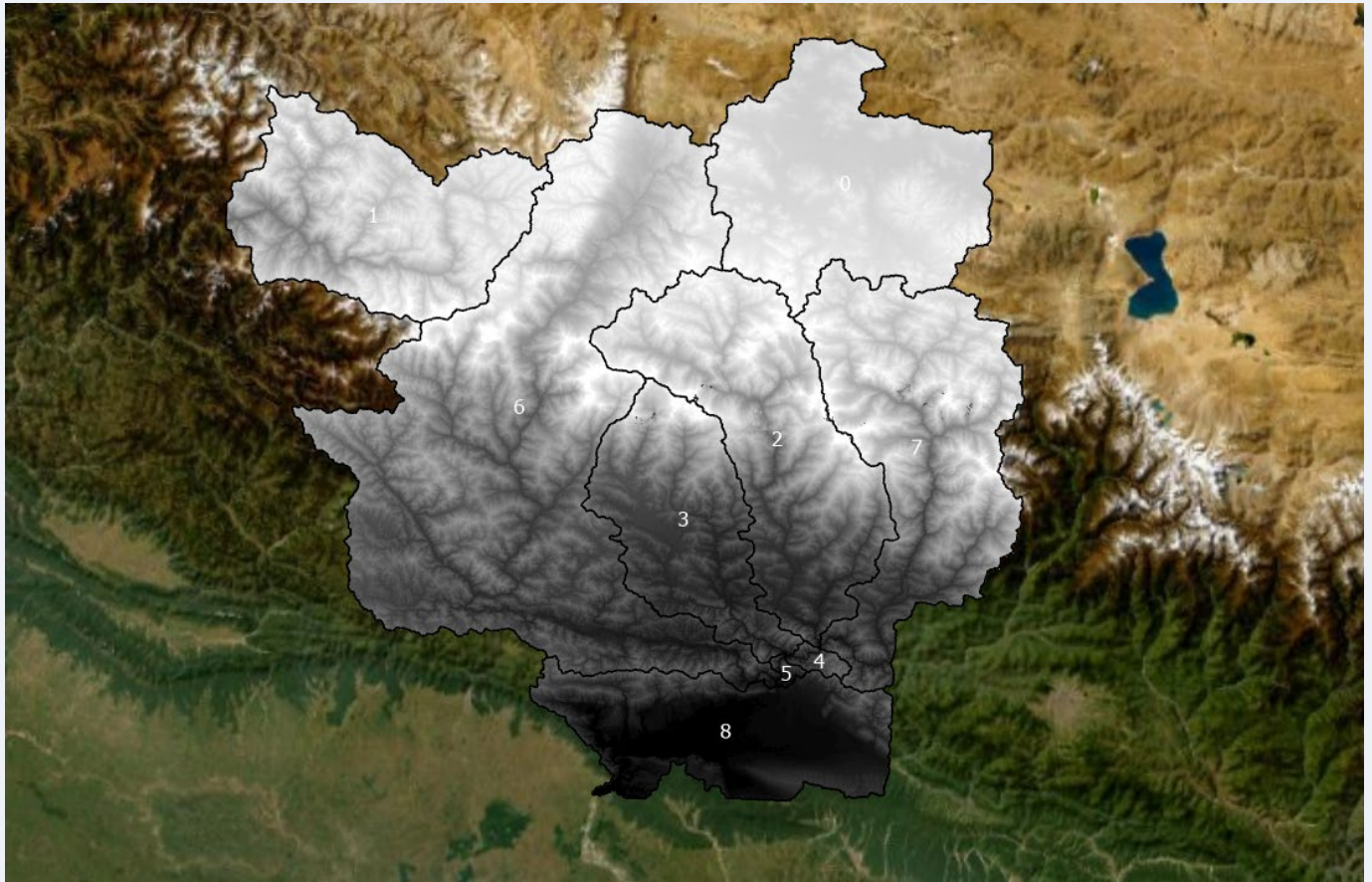


Figure 3. Watershed delineation map showing 9 sub-basins displayed over a shaded-relief DEM.

4.2 Rainfall Frequency Analysis

Annual maximum daily rainfall data for 1990-2024 was extracted from the CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) dataset (Funk *et al.*, 2015) using Google Earth Engine. The data was spatially averaged over the watershed boundary to obtain representative areal rainfall depths.

Flood frequency analysis was performed using the Log-Pearson Type III (LP3) distribution in EasyFit software,

following the guidelines of Bulletin 17C (England *et al.*, 2019). The LP3 distribution is the standard recommended method for frequency analysis in hydrological studies and is widely applied across South Asian river basins. Goodness-of-fit was assessed.

Design rainfall depths were extracted from the fitted LP3 distribution for six return periods using the inverse cumulative distribution function at the non-exceedance probabilities listed in Table 4.

Table 4. Return periods and corresponding probability values used in frequency analysis.

Return Period (yr)	Non-exceedance P
2	0.50
5	0.80
10	0.90
25	0.96
50	0.98
100	0.99

4.3 SCS Curve Number Derivation

4.3.1 Land Cover Classification

Land cover data was obtained from ESA WorldCover 2021 at 10 m resolution (ESA, 2021). The WorldCover product provides 11 land cover classes derived from Sentinel-1 and Sentinel-2 satellite imagery and was processed and clipped to the watershed boundary in Google Earth Engine.

4.3.2 Hydrologic Soil Group Determination

Soil hydrologic groups (HSG) were derived from the OpenLandMap soil texture classification at 250 m resolution. USDA soil texture classes were reclassified to the four NRCS Hydrologic Soil Groups (A, B, C, D) based on infiltration capacity, following USDA-NRCS National Engineering Handbook guidance (USDA-NRCS, 2004):

Table 5. USDA soil texture to HSG reclassification.

USDA Texture Class	Code	HSG	Infiltration Rate
Clay, silty clay, sandy clay	1-3	D	Very low (<1.27 mm/hr)
Clay loam, silty clay loam, sandy clay loam	4-6	C	Low (1.27-3.81 mm/hr)
Loam, silty loam, sandy loam, silt	7-10	B	Moderate (3.81-7.62 mm/hr)
Loamy sand, sand	11-12	A	High (>7.62 mm/hr)

Analysis revealed that the Narayani basin is dominated by HSG B and HSG C, consistent with the mixed geology of Himalayan hill soils and Terai alluvial deposits.

4.3.3 CN Lookup and Spatial Analysis

A composite CN raster was generated in Google Earth Engine by combining the WorldCover LULC and HSG rasters using a pixel-level lookup approach. A combined key was computed as:

$$\text{Key} = \text{LULC_code} \times 10 + \text{HSG_code} \quad (1)$$

Each unique key was mapped to its corresponding CN value from the NRCS TR-55 standard curve number table (USDA-NRCS, 2004). Area-weighted composite CN values for each sub-basin were calculated using zonal statistics (mean reducer) at 100 m resolution.

Table 6. SCS Curve Number lookup table for HSG B and C (dominant groups in the Narayani basin).

WorldCover Class	Land Cover	HSG B	HSG C
10	Tree cover (forest)	60	73
20	Shrubland	56	70
30	Grassland	61	74
40	Cropland	78	85
50	Built-up	85	90
60	Bare / sparse vegetation	86	91
80	Water bodies	100	100
90	Herbaceous wetland	84	88

4.4 HEC-HMS Rainfall-Runoff Modeling

4.4.1 Basin Model Setup

The HEC-HMS (USACE, 2016) basin model was constructed with 9 sub-basin elements connected by 4 reach elements, routing to a single outlet (Sink-1). The SCS Curve Number loss method was applied for all sub-basins, and the SCS Unit Hydrograph method was used for runoff transformation.

4.4.2 Time of Concentration and Lag Time

Time of concentration (T_c) for each sub-basin was calculated using the SCS formula adapted for SI units:

$$T_c = (L^{0.8} \times ((1000/CN) - 9)^{0.7}) / (441 \times Y^{0.5}) \quad (2)$$

where L is the longest flow-path length (m), S is the maximum potential retention (mm), and Y is the average basin slope (%). SCS lag time and the retention and initial abstraction parameters were computed as:

$$Lag (min) = 0.6 \times Tc \times 60 \quad (3)$$

$$S = (25400/CN) - 254 \quad (4)$$

$$Ia (mm) = 0.2 \times S \quad (5)$$

4.4.3 Reach Routing

Muskingum routing was applied to all 4 reaches. Travel time K was estimated as the ratio of reach length to flood-wave celerity (assumed 1.5 m/s for Himalayan rivers). The weighting factor X was set to 0.2 for all reaches, representing moderate attenuation typical of natural river channels. The number of sub-reaches was set equal to K (hours) divided by the computation time step (1 hour) for numerical stability.

4.4.4 Meteorological Model

4.5 HEC-RAS 2D Hydraulic Modeling

4.5.1 Model Domain

The HEC-RAS 2D (Brunner, 2016) model domain was defined as a 4 km buffer polygon around the 37.6 km study reach of the Narayani River, from the gorge exit near Devghat to downstream of Narayanghat. The buffer was generated in ArcGIS Pro using the river centerline shapefile and represents the potential flood inundation zone.

4.5.2 Terrain

The ALOS PALSAR DEM at 12.5 m resolution was loaded as the primary terrain in RAS Mapper. A single DEM tile covering the study reach was used to avoid tile-boundary

The HEC-HMS meteorological model was configured using the Hypothetical Storm method with an SCS Type II temporal distribution and a 24-hour storm duration. Design storm depths from the Log-Pearson III frequency analysis were applied uniformly across all sub-basins, and six simulation runs were executed, one for each return period.

4.4.5 Control Specifications

Simulations were run from 01 January 2000 00:00 to 10 January 2000 00:00 with a 1-hour computation interval. The 10-day simulation window was selected to capture the full hydrograph, including the rising limb, peak, and complete recession, for the largest (100-year) design storm.

artifacts that can cause artificial flow discontinuities in multi-tile mosaics.

4.5.3 Computational Mesh

A 2D computational mesh was generated within the model domain at 30 m nominal cell size to adequately resolve the braided channel morphology and bank features of the Narayani River. The mesh comprised approximately 333,000 cells, providing sufficient resolution to capture channel-floodplain interactions.

4.5.4 Manning's Roughness Coefficients

Spatially distributed Manning's roughness coefficients were assigned using the ESA WorldCover 2021 land cover raster (ESA, 2021) as a calibration layer in RAS Mapper, following the class assignments in Table 7.

Table 7. Manning's roughness coefficients assigned by land cover class.

WorldCover Class	Land Cover	Manning's n
10	Tree cover (dense forest)	0.10
20	Shrubland	0.07
30	Grassland	0.05
40	Cropland	0.04
50	Built-up / urban	0.03
60	Bare / sparse vegetation	0.02
80	Water bodies	0.03
90	Herbaceous wetland	0.06

4.5.5 Boundary Conditions

Two boundary condition lines were defined within the 2D flow area perimeter:

- Upstream boundary (BCLine: US) — flow hydrograph imported from the HEC-HMS DSS output file (Sink-1, FLOW time series), representing the design flood hydrograph for each return period. An energy grade

slope of 0.00125 was specified for flow distribution along the boundary line.

- Downstream boundary (BCLine: DS) — normal-depth condition with a friction slope of 0.00125, calculated from the DEM elevation difference (121 m upstream to 76 m downstream) divided by the reach length (35,957 m).

4.5.6 Simulation Parameters

Unsteady flow simulations were executed with a 1-minute computation interval and a 1-hour output interval. Results

5. RESULTS AND DISCUSSION

5.1 Watershed Characteristics

The watershed delineation produced 9 sub-basins with varying morphometric characteristics reflecting the diverse

were mapped at 1-hour intervals throughout the 10-day simulation window for each return period.

physiographic zones of the Narayani basin. The largest sub-basin (Sub-basin 4) covers 7,636.4 km² in the upper watershed, while the smallest (Sub-basin 9) covers only 66.1 km². Basin slopes range from 0.007 m/m (Sub-basin 2) to 0.029 m/m (Sub-basin 7), reflecting the steeper terrain of the northern sub-basins.

Table 8. Sub-basin morphometric characteristics and computed hydrological parameters.

Sub-basin	Area (km ²)	Longest Flow Path (km)	Basin Slope (m/m)	Tc (hr)	Lag (min)
Sub-basin 1	2,234.5	211.70	0.026	80.64	2,903
Sub-basin 2	166.5	43.83	0.007	51.57	1,857
Sub-basin 3	1,342.6	89.78	0.011	70.19	2,527
Sub-basin 4	7,636.4	238.66	0.025	116.45	4,192
Sub-basin 5	1,848.9	93.10	0.015	64.62	2,326
Sub-basin 6	5,198.8	194.68	0.027	110.10	3,964
Sub-basin 7	1,994.2	97.04	0.029	49.09	1,767
Sub-basin 8	2,937.8	149.35	0.026	75.11	2,704
Sub-basin 9	66.1	22.19	0.026	15.34	552

5.2 Rainfall Frequency Analysis Results

The Log-Pearson Type III distribution provided the best fit to the annual maximum daily rainfall series. The fitted

distribution parameters and goodness-of-fit statistics indicated an acceptable fit to the observed CHIRPS data (*Funk et al., 2015*). Design rainfall depths for the six return periods are presented in Table 9.

Table 9. Design rainfall depths from Log-Pearson III frequency analysis.

Return Period (yr)	P Value	Design Rainfall Depth (mm)
2	0.50	93.66
5	0.80	123.05
10	0.90	145.83
25	0.96	178.62
50	0.98	206.13
100	0.99	236.45

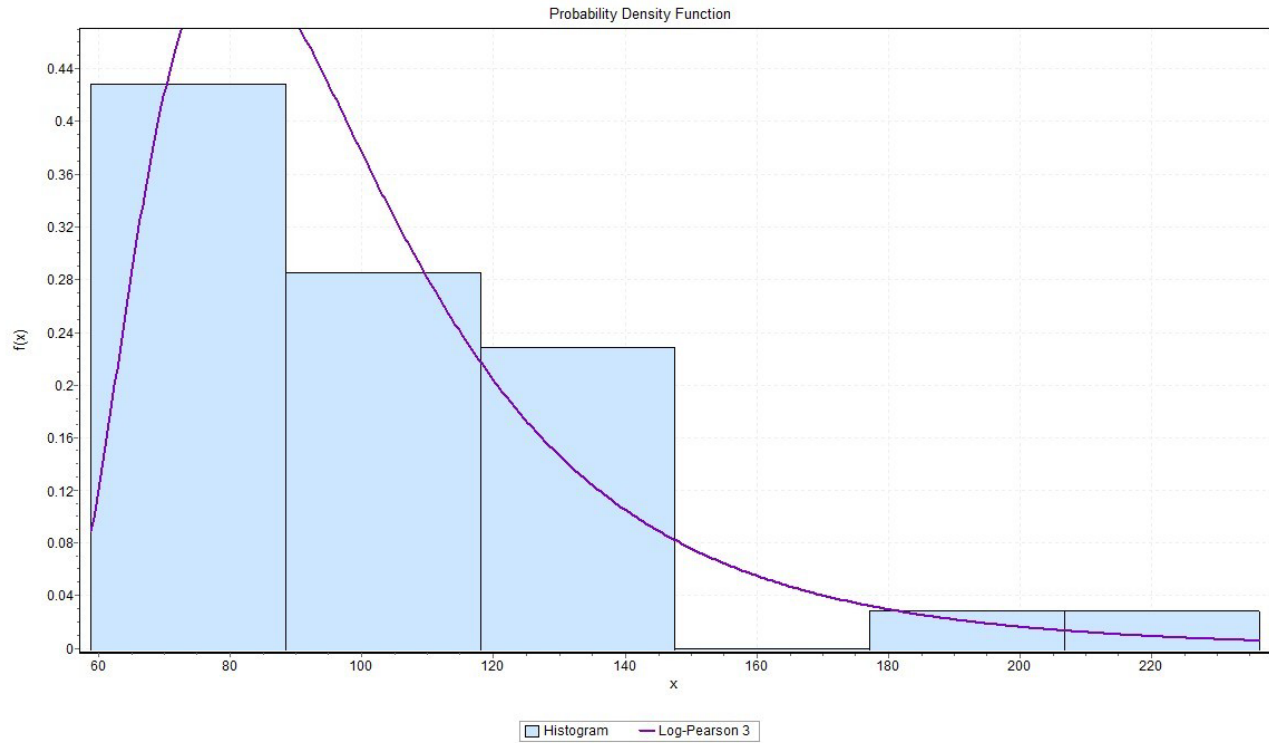


Figure 4. Log-Pearson III fitted frequency curve showing annual maximum rainfall versus return period, with confidence intervals and observed data points (exported from EasyFit).

5.3 SCS Curve Number Results

Land cover analysis of the floodplain study area revealed a dominance of tree cover (LULC class 10) and bare/sparse vegetation (class 60) in the upper watershed, transitioning

to cropland (class 40) and built-up areas (class 50) in the lower Terai zone. Soil analysis confirmed HSG B and C as the dominant hydrologic soil groups, consistent with the mixed loamy and clay-loam soils characteristic of the Nepal Middle Hills and Terai.

Table 10. SCS Curve Number and initial abstraction values for each sub-basin.

Sub-basin	CN (GEE)	CN (ArcGIS Zonal)	Ia (mm)	S (mm)
Sub-basin 1	71	70.39	21.37	106.86
Sub-basin 2	66	65.66	26.57	132.85
Sub-basin 3	66	65.99	26.18	130.89
Sub-basin 4	61	61.17	32.24	161.21
Sub-basin 5	65	65.05	27.29	136.44
Sub-basin 6	56	55.51	40.71	203.54
Sub-basin 7	64	63.52	29.17	145.87
Sub-basin 8	63	63.06	29.75	148.76
Sub-basin 9	65	65.10	27.23	136.17

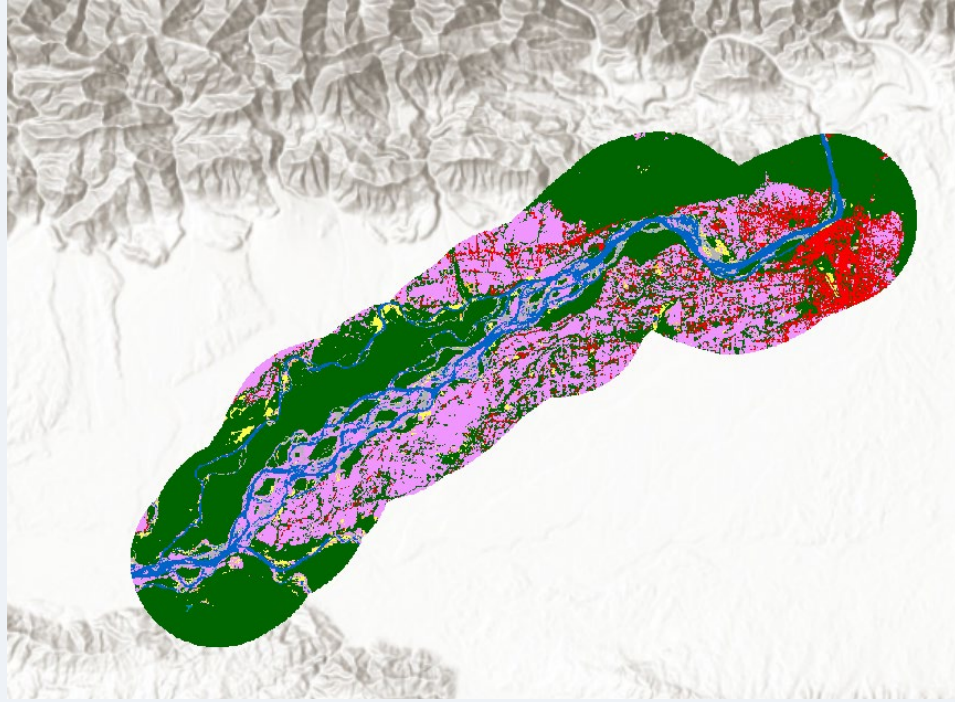


Figure 5. Spatially distributed CN raster map over the watershed, showing CN values ranging from 55-100 with sub-basin boundaries and legend (exported from GEE / ArcGIS Pro).

5.4 HEC-HMS Simulation Results

HEC-HMS simulations for the six design storm events produced flood hydrographs at the watershed outlet (Sink-

1). Peak discharge increased consistently with return period, ranging from 1,374.9 m³/s for the 2-year event to 8,551.5 m³/s for the 100-year event, confirming the physical plausibility of the model.

Table 11. HEC-HMS simulation results for six return periods at the watershed outlet (Sink-1).

Return Period (yr)	Rainfall (mm)	Peak Discharge (m ³ /s)	Time of Peak	Volume (mm)
2	93.66	1,374.9	5 Jan 2000, 03:00	17.34
5	123.05	2,562.1	5 Jan 2000, 03:00	32.75
10	145.83	3,620.6	5 Jan 2000, 04:00	46.56
25	178.62	5,296.5	5 Jan 2000, 04:00	68.51
50	206.13	6,806.0	5 Jan 2000, 04:00	88.33
100	236.45	8,551.5	5 Jan 2000, 04:00	111.30

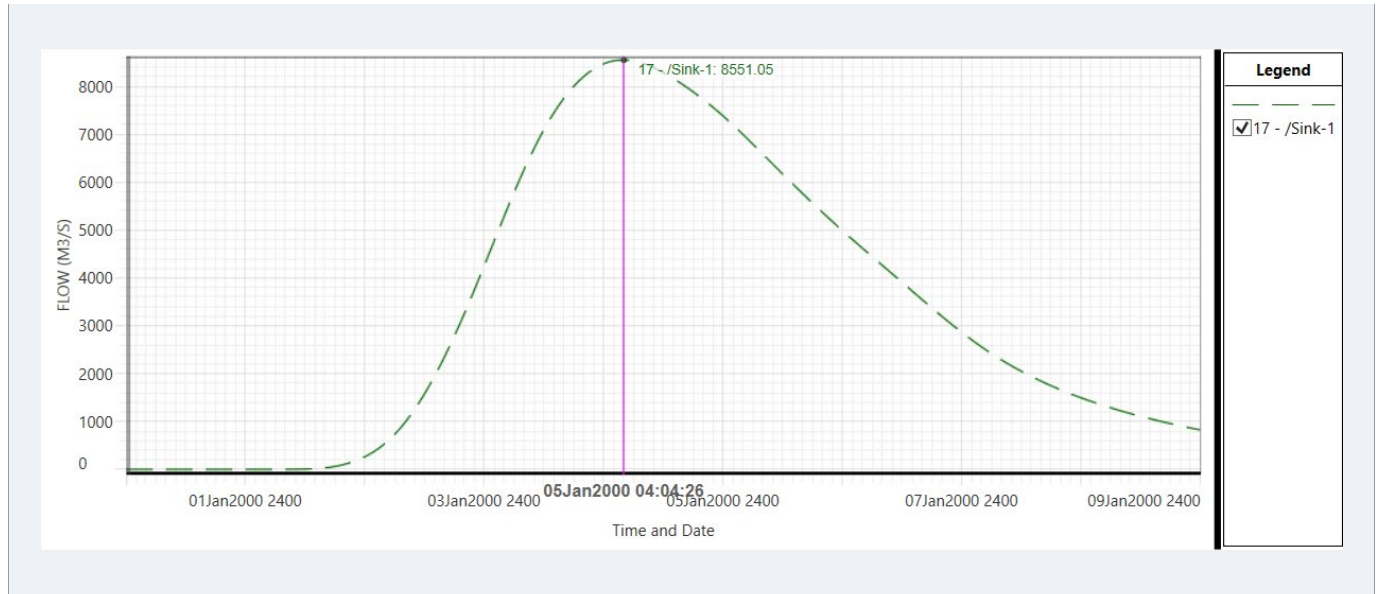


Figure 6. Flood hydrographs at Sink-1 for 100 year return period, plotted on a graph of discharge (m^3/s) versus time.

The 100-year design storm produced a peak discharge of $8,551.5 \text{ m}^3/\text{s}$ at Sink-1, occurring approximately 4 days after storm onset. This lag reflects the long travel times in the large upper sub-basins (Sub-basin 4 lag = 4,192 minutes) and the Muskingum routing attenuation through the 4 reaches. The peak discharge of $8,551.5 \text{ m}^3/\text{s}$ for the 100-year return period is consistent with published estimates for the Narayani basin at Narayanghat, which report mean annual flood values in the range of $8,000\text{--}12,000 \text{ m}^3/\text{s}$ (DHM, 2020).

5.5 HEC-RAS 2D Simulation Results

The HEC-RAS 2D unsteady-flow simulation for the 100-year return period produced spatially distributed results for flood depth, water surface elevation, and flow velocity throughout the 37.6 km study reach. Maximum inundation occurred at approximately 5 January 2000, 04:00, coinciding with the peak discharge from HEC-HMS. These coupled rainfall-runoff and 2D hydrodynamic results are consistent in approach with comparable coupled 1D/2D flood-mapping studies applied to flood-prone South Asian floodplains (Patel *et al.*, 2017).

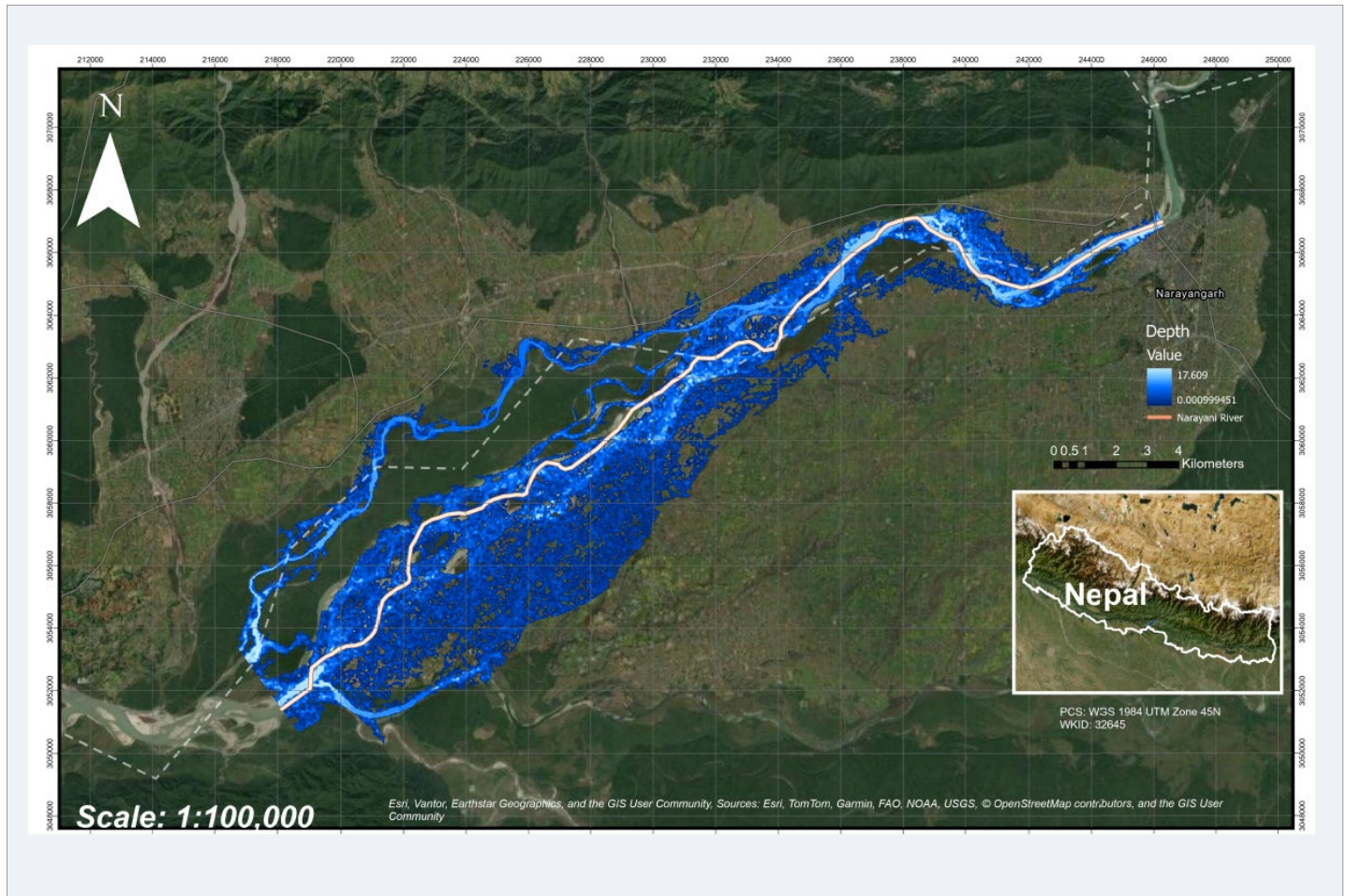


Figure 7. HEC-RAS 2D flood depth map at peak time (05 Jan 2000, 04:00) for the 100-year return period, classified as 0-0.5 m, 0.5-1.5 m, 1.5-3 m, and >3 m, with river channel, settlement boundaries, and scale bar.

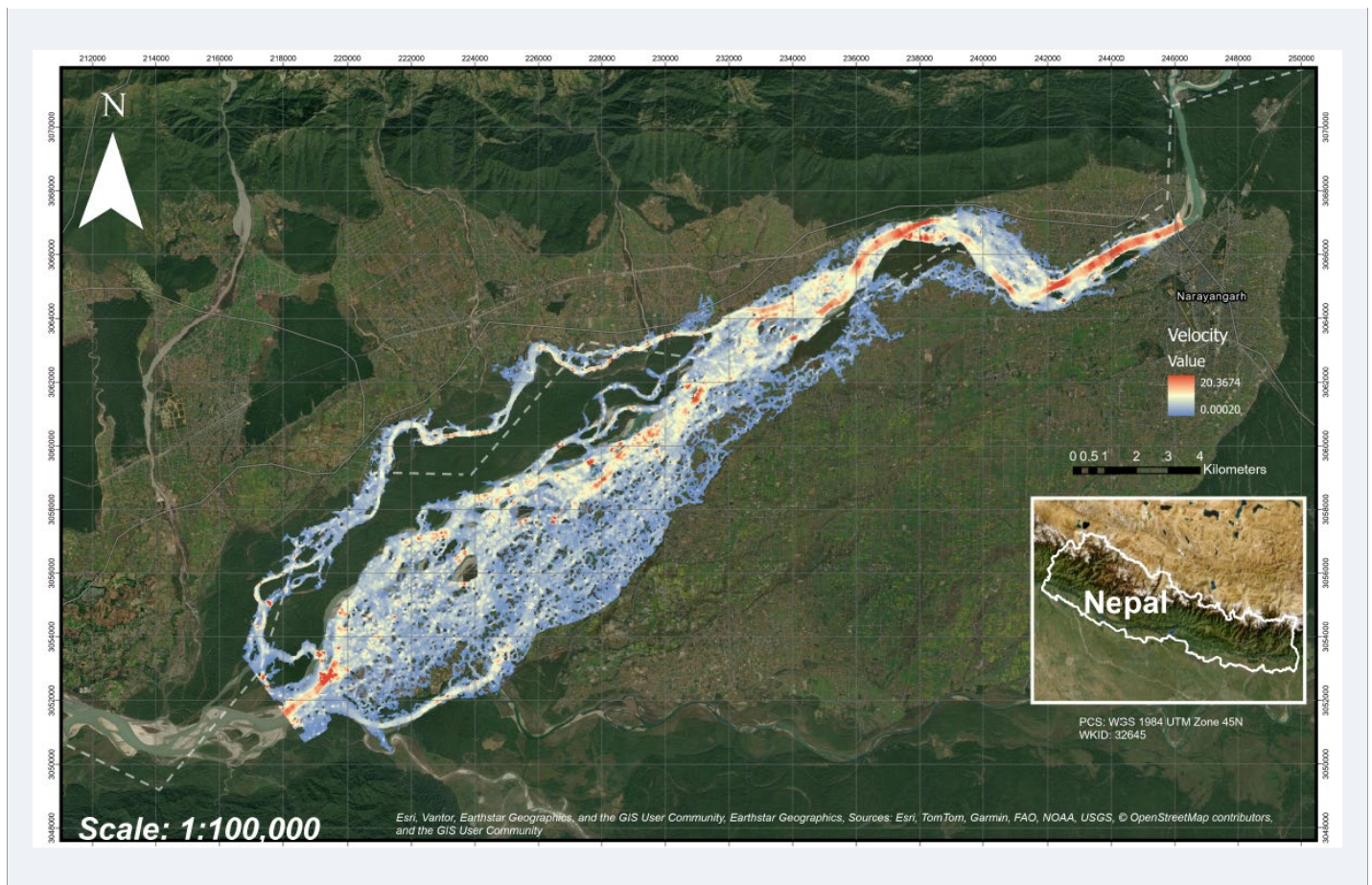


Figure 8. HEC-RAS 2D flow velocity map at peak time for the 100-year return period, showing velocity (m/s) with high-velocity zones highlighted in the channel.

5.6 Flood Inundation Maps

Flood inundation maps were produced for all six return periods, showing the spatial extent and depth of flooding

within the study reach. The maps reveal a clear relationship between return period and flood extent, with the 100-year event inundating significantly larger areas than the 2-year event.

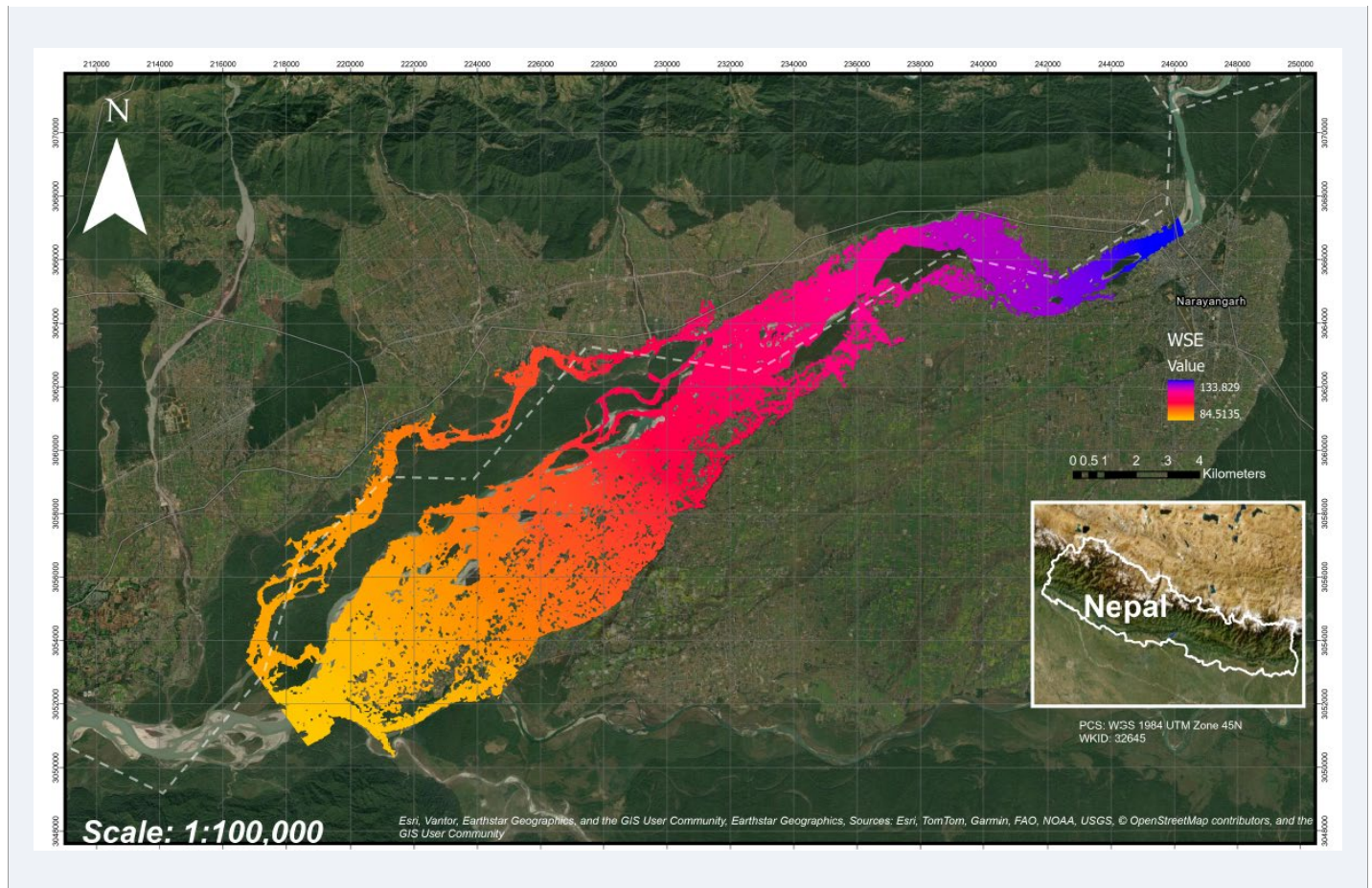


Figure 9. Multi-panel flood inundation extent map showing all six return periods (2, 5, 10, 25, 50, 100 year) at peak time, with a consistent color scale, legend, north arrow, and scale bar.

The flood inundation analysis reveals that the main channel of the Narayani River experiences depths exceeding 10-15 m during the 100-year event within the gorge section, reflecting the confined nature of the channel in this reach.

Downstream of the gorge exit, floodwaters spread onto the Terai floodplain, with depths of 1-5 m affecting agricultural lands and settlements in Chitwan district.

6. CONCLUSION

This study successfully developed a comprehensive flood inundation mapping framework for the Narayani River basin, Nepal, using an integrated hydrological and hydrodynamic modeling approach. The key conclusions are:

- The Log-Pearson III distribution provided an acceptable fit to CHIRPS annual maximum rainfall data (Funk *et al.*, 2015) for 1990-2024, yielding design storm depths ranging from 93.66 mm (2-year) to 236.45 mm (100-year) for the Narayani basin.
- SCS Curve Numbers derived from ESA WorldCover 2021 (ESA, 2021) and OpenLandMap soil data ranged from 56 to 71 across the 9 sub-basins, reflecting the

predominantly forested and agricultural land cover on HSG B and C soils.

- HEC-HMS rainfall-runoff simulations (USACE, 2016) produced physically plausible peak discharge estimates ranging from 1,374.9 m³/s (2-year) to 8,551.5 m³/s (100-year) at the watershed outlet, consistent with published flood estimates for the Narayani basin (DHM, 2020).
- HEC-RAS 2D unsteady-flow modeling (Brunner, 2016) successfully simulated flood inundation for all six return periods, producing spatially distributed maps of flood depth, water surface elevation, and flow velocity.
- The integrated GEE-HMS-RAS modeling framework demonstrated the feasibility of producing flood inundation maps using freely available remote sensing data, making it applicable to data-scarce river basins in developing countries.

- The study provides critical flood hazard information for the Narayani floodplain that can support flood risk assessment, land-use planning, evacuation route

7. LIMITATIONS AND FUTURE WORK

7.1 Limitations

- The absence of observed discharge gauge data at the watershed outlet precluded formal model calibration and validation. Future studies should incorporate DHM gauge records for calibration.
- ERA5 reanalysis discharge data was found to significantly overestimate peak flows in the Himalayan basin ($>80,000 \text{ m}^3/\text{s}$ versus a realistic $\sim 10,000 \text{ m}^3/\text{s}$) and was excluded from the analysis. Improved discharge datasets from physical gauging stations are needed.
- The SCS Curve Number method assumes a uniform storm over the entire watershed, which may not accurately represent the orographic rainfall patterns of the Himalayan basin.
- HEC-RAS 2D simulation was limited to a 37.6 km study reach due to computational constraints; the downstream floodplain extending to Narayanghat and beyond was not fully captured.
- Manning's roughness coefficients were assigned from land cover class averages reported in the literature and were not field-calibrated for the Narayani floodplain.

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- The ALOS PALSAR DEM at 12.5 m resolution, while adequate for basin-scale modeling, may not capture fine-scale channel morphology and bank features critical for accurate inundation mapping.
- ### 7.2 Future Work
- Calibration and validation of HEC-HMS and HEC-RAS models using historical flood records and DHM gauge data
 - Integration of climate change projections to assess future flood frequency under different emission scenarios
 - Extension of the HEC-RAS 2D model domain to cover the full Narayani floodplain, including the Narayanghat urban area
 - High-resolution UAV or LiDAR survey of critical channel cross-sections for improved terrain representation
 - Population exposure and economic damage assessment using flood depth-damage functions
 - Real-time flood forecasting system integration using the HMS-RAS modeling chain with NWP rainfall forecasts
- city by coupled 1D/2D hydrodynamic modeling: a case application of the new HEC-RAS 5. *Natural Hazards*, 89(1), 93–130.
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- Yamazaki, D., Ikeshima, D., Sosa, J., Bates, P. D., Allen, G. H., & Pavelsky, T. M. (2019). MERIT Hydro: A high-resolution global hydrography map based on latest topography dataset. *Water Resources Research*, 55(6), 5053–5073.

APPENDIX A. GEE CODE — SCS CURVE NUMBER MAPPING

The following Google Earth Engine JavaScript code was used to derive SCS Curve Numbers for the Narayani basin:

```
var subbasin = ee.FeatureCollection('projects/ee-abhishekkarki520/assets/subbasin');
```

```
var worldcover = ee.ImageCollection('ESA/WorldCover/v200').first().select('Map').clip(subbasin);
var soilTexture = ee.Image('OpenLandMap/SOL/SOL_TEXTURE-CLASS_USDA-TT_M/v02').select('b0').clip(subbasin);
var hsg = soilTexture.remap([1,2,3,4,5,6,7,8,9,10,11,12],[4,4,4,3,3,3,2,2,2,2,1,1],0).rename('HSG');
var combined = worldcover.multiply(10).add(hsg).rename('key');
```

APPENDIX B. EXCEL CALCULATION — TC AND LAG TIME

Table A1. Time of concentration and lag time calculations using the SCS formula (SI units, constant = 441).

Sub-basin	CN	L (km)	Y (m/m)	L (m)	Y (%)	Tc (hr)	Lag (min)
Sub-basin 1	70.39	211.70	0.026	211,695	2.65	80.64	2,903
Sub-basin 2	65.66	43.83	0.007	43,830	0.67	51.57	1,857
Sub-basin 3	65.99	89.78	0.011	89,781	1.12	70.19	2,527
Sub-basin 4	61.17	238.66	0.025	238,662	2.49	116.45	4,192
Sub-basin 5	65.05	93.10	0.015	93,099	1.47	64.62	2,326
Sub-basin 6	55.51	194.68	0.027	194,680	2.68	110.10	3,964
Sub-basin 7	63.52	97.04	0.029	97,045	2.94	49.09	1,767
Sub-basin 8	63.06	149.35	0.026	149,354	2.57	75.11	2,704
Sub-basin 9	65.10	22.19	0.026	22,194	2.62	15.34	552

APPENDIX C. MUSKINGUM ROUTING PARAMETERS

Table A2. Muskingum routing parameters for each reach (wave celerity = 1.5 m/s).

Reach	Length (km)	K (hr)	X	Sub-reaches
Reach 1	37.23	7	0.2	7
Reach 2	18.80	4	0.2	4
Reach 3	53.49	10	0.2	10
Reach 4	152.24	28	0.2	28

```

# =====
# FLOOD INUNDATION MAPPING – NARAYANI RIVER
# Animation Code to animate the frames from Hec-RAS
# =====

import os, h5py, numpy as np, matplotlib.pyplot as plt, matplotlib.tri as tri
import matplotlib.animation as animation
from matplotlib.lines import Line2D
from matplotlib.scalebar.scalebar import ScaleBar
from datetime import datetime, timedelta
import rasterio, contextily as cx
from rasterio.plot import reshape_as_image
from matplotlib.colors import LinearSegmentedColormap, LightSource
import imageio, imageio_ffmpeg
import shutil, glob

# ===== USER SETTINGS =====
HDF_GEOM = r"C:\Users\Lenovo\Documents\FIM\FIM.g01.hdf"
HDF_RESULTS = r"C:\Users\Lenovo\Documents\FIM\FIM.p01.hdf"
TERRAIN_TIF = r"C:\Users\Lenovo\Documents\FIM\Terrain\Terrain.AP_16653_FBS_F0540_RT1.dem.tif"
LANDCOVER_TIF = r"C:\Users\Lenovo\Documents\FIM\Land Classification\LandCover.tif"
AREA_NAME = "Narayani"
OUTPUT_MP4 = "Narayani_Inundation_Labeled.mp4"

NUM_FRAMES = 60
FPS = 10
FIG_SIZE = (16.54, 11.69) # A3 landscape
DPI = 150
CRS = "EPSG:32645"

# ===== VIBRANT BLUE WATER =====
colors = ["#e0f3ff", "#a6d7ff", "#6bb5ff", "#2e8ed6", "#0a4a7a", "#001f3f"]
cmap_water = LinearSegmentedColormap.from_list("BlueVibrant", colors, N=256)

# ===== 1. LOAD GEOMETRY =====
print("Loading geometry...")
with h5py.File(HDF_GEOM, 'r') as g:
    base = f'/Geometry/2D Flow Areas/{AREA_NAME}'
    coords = g[f'{base}/Cells Center Coordinate'][:]
    cell_elev = g[f'{base}/Cells Minimum Elevation'][:]
    perimeter = g[f'{base}/Perimeter'][:]

    bcs = []
    if 'Boundary Condition Lines' in g['/Geometry']:
        bc_group = g['/Geometry/Boundary Condition Lines']
        attrs = bc_group['Attributes'][:]
        info = bc_group['Polyline Info'][:]
        pts = bc_group['Polyline Points'][:]
        for i, attr in enumerate(attrs):
            name = attr[0].decode('utf-8')
            start, count = info[i][0], info[i][1]
            bcs.append({'name': name, 'points': pts[start:start+count]})

# ===== 2. LOAD RESULTS & TIMESTAMPS =====
with h5py.File(HDF_RESULTS, 'r') as p:
    ts = p['Results/Unsteady/Output/Output Blocks/Base Output/Unsteady Time Series']
    raw_ts = ts['Time Date Stamp'][:]
    if hasattr(raw_ts, 'flat'): raw_ts = raw_ts.flat[0]
    if isinstance(raw_ts, bytes): raw_ts = raw_ts.decode('utf-8')
    elif not isinstance(raw_ts, str): raw_ts = str(raw_ts)
    ref_dt = datetime.strptime(raw_ts, '%d%b%Y %H:%M:%S')

    time_hours = ts['Time'][:]
    all_times = [ref_dt + timedelta(hours=float(h)) for h in time_hours]
    wse_all = p[f'{ts.name}/2D Flow Areas/{AREA_NAME}/Water Surface'][:]

total = len(all_times)
idx = np.linspace(0, total-1, NUM_FRAMES, dtype=int) if total > NUM_FRAMES else np.arange(total)
timestamps = [all_times[i] for i in idx]
wse_data = wse_all[idx, :]

# ---- Depth max (full range) ----
depth_full = wse_data - cell_elev
depth_vmax = np.nanmax(depth_full)
print(f"Full depth range: 0 – {depth_vmax:.2f} m")

# ===== 3. TERRAIN HILLSHADE (very faint) =====
hillshade = None; dem_extent = None
if os.path.exists(TERRAIN_TIF):
    with rasterio.open(TERRAIN_TIF) as dem:
        dem_data = dem.read(1)
        dem_extent = [dem.bounds.left, dem.bounds.right, dem.bounds.bottom, dem.bounds.top]
        ls = LightSource(azdeg=315, altdeg=45)
        hillshade = ls.hillshade(dem_data, vert_exag=2.0)

# ===== 4. LANDCOVER (very faint) =====
landcover_layer = None; lc_extent = None; lc_cmap = None
if os.path.exists(LANDCOVER_TIF):
    with rasterio.open(LANDCOVER_TIF) as lc:
        n_bands = lc.count
        if n_bands >= 3:
            lc_data = lc.read([1,2,3])
            landcover_layer = reshape_as_image(lc_data)
        else:
            lc_band = lc.read(1).astype(np.float32)
            nodata = lc.nodata
            if nodata is not None: lc_band[lc_band == nodata] = np.nan
            valid = lc_band[~np.isnan(lc_band)]
            vmin, vmax = (np.nanmin(valid), np.nanmax(valid)) if len(valid)>0 else (0,1)
            if vmax > vmin: lc_band = (lc_band - vmin) / (vmax - vmin)

```



```

landcover_layer = lc_band
lc_cmap = 'tab20'
lc_extent = [lc.bounds.left, lc.bounds.right, lc.bounds.bottom, lc.bounds.top]

# ===== 5. TRIANGULATION =====
triang = tri.Triangulation(coords[:,0], coords[:,1])

# ===== 6. FIGURE =====
fig, ax = plt.subplots(figsize=FIG_SIZE, facecolor='black')
plt.subplots_adjust(left=0.05, right=0.95, top=0.92, bottom=0.08)
ax.set_facecolor('#1a1a2e')
ax.set_aspect('equal')
ax.tick_params(colors='white', labelsize=10)
for spine in ax.spines.values():
    spine.set_edgecolor('white')
    spine.set_linewidth(1.2)
ax.set_xlabel("Easting (m)", color='white', fontsize=11)
ax.set_ylabel("Northing (m)", color='white', fontsize=11)

xmin, xmax = perimeter[:,0].min(), perimeter[:,0].max()
ymin, ymax = perimeter[:,1].min(), perimeter[:,1].max()
ax.set_xlim(xmin - 0.02*(xmax-xmin), xmax + 0.02*(xmax-xmin))
ax.set_ylim(ymin - 0.02*(ymax-ymin), ymax + 0.02*(ymax-ymin))

# ===== 7. SATELLITE BASEMAP =====
print("🌀 Adding satellite basemap...")
sat_ok = False
for provider, url, zoom in [
    ("ESRI", "https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/tile/{z}/{y}/{x}", 15),
    ("Google", "https://mt1.google.com/vt/lyrs=s&x={x}&y={y}&z={z}", 16)
]:
    if sat_ok: break
    try:
        cx.add_basemap(ax, crs=CRS, source=url, zoom=zoom, alpha=1.0, zorder=0)
        print(f"✅ {provider} loaded")
        sat_ok = True
    except Exception as e:
        print(f"⚠️ {provider} failed: {e}")
if not sat_ok:
    print("❌ No satellite – using plain dark background.")

# Hillshade (barely visible)
if hillshade is not None:
    ax.imshow(hillshade, cmap='gray', extent=dem_extent, alpha=0.08, zorder=1)

# Landcover (barely visible)
if landcover_layer is not None:
    if lc_cmap:
        ax.imshow(landcover_layer, cmap=lc_cmap, extent=lc_extent, alpha=0.03, zorder=2, interpolation='nearest')
    else:
        ax.imshow(landcover_layer, extent=lc_extent, alpha=0.03, zorder=2)

# ===== PERIMETER & BOUNDARY CONDITIONS =====
# Perimeter line
ax.plot(perimeter[:,0], perimeter[:,1], color='cyan', lw=1.5, ls='--', zorder=3)

# Boundary condition lines & labels
bc_color = '#FF1493'
for bc in bcs:
    pts = bc['points']
    ax.plot(pts[:,0], pts[:,1], color=bc_color, lw=2.5, zorder=3)
    # Label at the midpoint of the BC line
    mid_idx = len(pts) // 2
    x_mid, y_mid = pts[mid_idx, 0], pts[mid_idx, 1]
    # Offset label slightly so it doesn't sit exactly on the line
    offset_x = (xmax - xmin) * 0.01 # 1% of map width
    offset_y = (ymax - ymin) * 0.01
    ax.text(x_mid + offset_x, y_mid + offset_y,
            bc['name'], fontsize=9, color='white', fontweight='bold',
            ha='center', va='center',
            bbox=dict(facecolor=bc_color, alpha=0.7, boxstyle='round,pad=0.2', ec='none'))

# ===== LEGEND =====
legend_elements = [
    Line2D([0],[0], color=bc_color, lw=2.5, label='Boundary Condition'),
    Line2D([0],[0], color='cyan', lw=1.5, ls='--', label='2D Mesh Perimeter')
]
ax.legend(handles=legend_elements, loc='upper right', facecolor='white',
          framealpha=0.9, edgecolor='black', fontsize=9)

# Scale bar (auto-length)
map_width = xmax - xmin
target = map_width * 0.15
nice_scales = [1,2,5,10,25,50,100,250,500,1000,2000,5000,10000]
dx = min(nice_scales, key=lambda x: abs(x - target))
if dx < target * 0.5:
    idx_ = nice_scales.index(dx)
    if idx_ < len(nice_scales)-1:
        dx = nice_scales[idx_+1]
scalebar = ScaleBar(dx, units="m", location="lower right",
                    box_alpha=0.7, color='white', box_color='black',
                    font_properties={'size': 10})
ax.add_artist(scalebar)

# North arrow
ax.annotate('N', xy=(0.04,0.95), xytext=(0.04,0.88),
            arrowprops=dict(facecolor='white',width=4,headwidth=12),
            ha='center',va='center',fontsize=22,color='white',fontweight='bold',
            xycoords='axes fraction',textcoords='axes fraction')

```

```

# ===== ON-MAP TEXT =====
ax.text(0.5, 0.97, "Flood Inundation Mapping, Narayani River",
        transform=ax.transAxes, ha='center', va='top',
        fontsize=18, fontweight='bold', color='white',
        bbox=dict(facecolor='black', alpha=0.7, boxstyle='round,pad=0.3', ec='none'))

ax.text(0.5, 0.92, "Unsteady Flow Simulation: Flood Depth Map",
        transform=ax.transAxes, ha='center', va='top',
        fontsize=14, color='white',
        bbox=dict(facecolor='black', alpha=0.6, boxstyle='round,pad=0.3', ec='none'))

proj_txt = ax.text(0.98, 0.08, "PCS: WGS 1984 UTM Zone 45N | TVD: HEC-RAS",
        transform=ax.transAxes, ha='right', va='bottom',
        fontsize=9, color='white',
        bbox=dict(facecolor='black', alpha=0.7, boxstyle='round,pad=0.3', ec='none'))

ax.text(0.5, 0.03, "GEE • easyFit • HEC-HMS • HEC-RAS • Python",
        transform=ax.transAxes, ha='center', va='bottom',
        fontsize=11, color='white',
        bbox=dict(facecolor='black', alpha=0.7, boxstyle='round,pad=0.3', ec='none'))

# ===== DEPTH LAYER (full range) =====
deptho = depth_full[0]
deptho[deptho < 0.05] = np.nan
tpc = ax.tricontour(triang, deptho, cmap=cmap_water, shading='gouraud',
                    vmin=0, vmax=depth_vmax, alpha=0.75, zorder=4)

cbar = fig.colorbar(tpc, ax=ax, pad=0.02, aspect=35)
cbar.set_label("Depth (m)", color='white', size=13)
cbar.ax.yaxis.set_tick_params(color='white', labels=10)
cbar.outline.set_edgecolor('white')
plt.setp(plt.getp(cbar.ax.axes, 'yticklabels'), color='white')

# ===== ANIMATION UPDATE =====
def animate(frame):
    d = depth_full[frame]
    d[d < 0.05] = np.nan
    tpc.set_array(d)
    return tpc,

# ===== SAVE MP4 =====
frame_dir = "frames_temp_labeled"
os.makedirs(frame_dir, exist_ok=True)
print(f"Saving {len(timestamps)} frames...")
for i in range(len(timestamps)):
    animate(i)
    fig.savefig(os.path.join(frame_dir, f"frame_{i:04d}.png"),
                facecolor='black', dpi=DPI, bbox_inches='tight', pad_inches=0.1)
    print(f" Frame {i+1}/{len(timestamps)}", end='\r')
print("\nAssembling video...")

try:
    writer = imageio.get_writer(OUTPUT_MP4, fps=FPS, codec='libx264',
                               quality=8, pixelformat='yuv420p')
    for fname in sorted(glob.glob(os.path.join(frame_dir, "frame_*.png"))):
        img = imageio.imread(fname)
        writer.append_data(img)
    writer.close()
    print(f"✅ Labeled MP4 saved: {OUTPUT_MP4}")
except Exception as e:
    print(f"⚠️ MP4 failed ({e}), falling back to GIF...")
    ani = animation.FuncAnimation(fig, animate, frames=len(timestamps),
                                  interval=1000/FPS, blit=False)
    writer_gif = animation.PillowWriter(fps=FPS)
    ani.save(OUTPUT_MP4.replace('.mp4', '.gif'), writer=writer_gif, dpi=DPI)
    print(f"✅ Labeled GIF saved: {OUTPUT_MP4.replace('.mp4', '.gif')}")

shutil.rmtree(frame_dir, ignore_errors=True)
plt.close(fig)
print("Done.")

```